MAT 130

CHAPTER 7.2: Integration by Parts

In this section, we will discuss an integration technique that is essentially an antiderivative formulation of the formula for differentiating a product of two functions.

The Product Rule and Integration by Parts

Basic Formula

If $f(x)$ and $g(x)$ are functions of *x*, then

$$\int f(x)g(x)\,dx = f(x)\int g(x)\,dx - \int \{\frac{d}{dx} f(x)\int g(x)\,dx\}dx$$

We write simply $\int (fg)\,dx = f\int g\,dx - \int \{\frac{df}{dx}\int g\,dx\}dx$

The application of this formula is known as **Integration by Parts.**

Useful strategy for the selection of f and g:

LIATE

Logarithmic**, I**nverse Trigonometric**, A**lgebraic**, T**rigonometric**, E**xponential

Example 1 Evaluate $\int x\cos x\,dx$.

Solution: Using formula, we can write

$$\int x\cos x\,dx = x\int \cos x\,dx - \int \{\frac{d}{dx}(x)\int \cos x\,dx\}dx$$

$$= x\sin x - \int \sin x dx = x\sin x + \cos x + c$$

Example 2 Evaluate $\int xe^x\,dx$.

Solution: Using formula, we can write

$$\int xe^x\,dx = x\int e^x\,dx - \int \{\frac{d}{dx}(x)\int e^x\,dx\}dx$$

$$= xe^x - e^x + c = (x-1)e^x + c$$

Example 3 Evaluate $\int \ln x\,dx$.

Solution: Using formula, we can write

$$\int \ln x\,dx = \ln x \int 1\,dx - \int \{\frac{d}{dx}(\ln x)\int 1\,dx\}dx$$

$$= x\ln x - \int (1/x)\,x\,dx = x\ln x - x + c = x(\ln x - 1) + c$$

Repeated Integration by Parts

Example 4 Evaluate $\int x^2 e^{-x}\,dx$.

Solution: Using formula, we can write

$$\int x^2 e^{-x}\,dx = x^2 \int e^{-x}\,dx - \int \{\frac{d}{dx}(x^2)\int e^{-x}\,dx\}dx$$

$$= -x^2 e^{-x} + 2\int x e^{-x}\,dx = -x^2 e^{-x} + 2[x\int e^{-x}\,dx + \int e^{-x}\,dx]$$

$$= -x^2 e^{-x} - 2xe^{-x} - 2e^{-x} + c = -(x^2 + 2x + 2)e^{-x} + c$$

Example 5 Evaluate $\int e^x \cos x\,dx$.

Solution: Let $I = \int e^x \cos x\,dx$. Then using formula, we can write

$$I = \cos x \int e^x\,dx - \int \{\frac{d}{dx}(\cos x)\int e^x\,dx\}dx$$

$$\Rightarrow I = e^x \cos x + \int e^x \sin x\,dx$$

$$\Rightarrow I = e^x \cos x + \sin x \int e^x\,dx - \int e^x \cos x\,dx]$$

$$\Rightarrow I = e^x \cos x + e^x \sin x - I + c$$

$$\Rightarrow 2I = e^x \cos x + e^x \sin x + c$$

Hence, $\int e^x \cos x\,dx = \frac{1}{2}e^x \cos x + \frac{1}{2}e^x \sin x + k$, where $k = c/2$.

Basic Formula

1. $\int e^{ax} \cos bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\cos bx + b\sin bx) + c$

2. $\int e^{ax} \sin bx\,dx = \frac{e^{ax}}{a^2 + b^2}(a\sin bx - b\cos bx) + c$

A Tabular Method for Repeated Integration by Parts

Tabular Integration by Parts

Step 1. Differentiate $f(x)$ repeatedly until you obtain 0, and list the results in the first column.

Step 2. Integrate $g(x)$ repeatedly and list the results in the second column.

Step 3. Draw an arrow from each entry in the first column to the entry that is one row down in the second column.

Step 4. Label the arrows with alternating + and − signs, starting with a +.

Step 5. For each arrow, form the product of the expressions at its tip and tail and then multiply that product by +1 or −1 in accordance with the sign on the arrow. Add the results to obtain the value of the integral.

The process is illustrated in the following table for the evaluation of the integral $\int (x^2 - x + 2)\cos x\,dx$:

Repeated differentiation	**Repeated integration**	**Product of the expressions**	**Multiply the product by**
$x^2 - x + 2$	$\cos x$		
$2x - 1$	$\sin x$	$(x^2 - x + 2)\sin x$	$+1$
2	$-\cos x$	$(2x-1)(-\cos x)$	-1
0	$-\sin x$	$(2)(-\sin x)$	$+1$

Therefore, using tabular method, we get

$$\int (x^2 - x + 2)\cos x = (x^2 - x + 2)\sin x - (2x-1)(-\cos x) - 2\sin x + c$$
$$= (x^2 - x)\sin x + (2x-1)\cos x + c$$

REDUCTION FORMULA

Integration by parts can be used to derive ***reduction formulas*** for integrals. These are formulas that express an integral involving a power of a function in terms of an integral that involves a *lower* power of that function.

Reduction formula of $\int \sin^m x \cos^n x\,dx$

Let $I_{m,n} = \int \sin^m x \cos^n x\,dx = \int \sin^m x \cos^{n-1} x \cos x\,dx$

$$= \cos^{n-1} x \int \sin^m x \cos x\,dx - \int \{\frac{d}{dx}(\cos^{n-1} x)\int \sin^m x \cos x\,dx\}\,dx$$

$$= \frac{\sin^{m+1} x \cos^{n-1} x}{m+1} + \frac{n-1}{m+1}\int \sin^{m+2} x \cos^{n-2} x\,dx$$

$$= \frac{\sin^{m+1} x \cos^{n-1} x}{m+1} + \frac{n-1}{m+1}\int \sin^m x \cos^{n-2} x(1-\cos^2 x)\,dx$$

$$= \frac{\sin^{m+1} x \cos^{n-1} x}{m+1} + \frac{n-1}{m+1}\int \sin^m x \cos^{n-2} x \; dx + \frac{n-1}{m+1}\int \sin^m x \cos^n x \; dx$$

$$I_{m,n} = \frac{\sin^{m+1} x \cos^{n-1} x}{m+n} + \frac{n-1}{m+n} I_{m,n-2} \text{ with } m+n \neq 0 \qquad (1)$$

Again, $I_{m,n} = \int \sin^m x \cos^n x\,dx = \int \cos^n x \sin^{m-1} x \sin x\,dx$

$$= \sin^{m-1} x \int \cos^n x \sin x\,dx - \int \{\frac{d}{dx}(\sin^{m-1} x)\int \cos^n x \sin x\,dx\}\,dx$$

$$I_{m,n} = -\frac{\sin^{m-1} x \cos^{n+1} x}{m+n} + \frac{m-1}{m+n} I_{m-2,n} \text{ with } m+n \neq 0 \qquad (2)$$

Reduction formula of $\int \cos^n x\,dx$

Putting $m=0$ in (1), we get

$$I_n = \int \cos^n x\,dx = \frac{\sin x \cos^{n-1} x}{n} + \frac{n-1}{n} I_{n-2} \text{ with } n \neq 0 \qquad (3)$$

Reduction formula of $\int \sin^m x\,dx$

Again, putting $n=0$ in (2), we get

$$I_m = \int \sin^m x\,dx = -\frac{\sin^{m-1} x \cos x}{m} + \frac{m-1}{m} I_{m-2} \text{ with } m \neq 0 \qquad (4)$$

Example 8 Evaluate $\int \cos^4 x\,dx$.

Solution: Using Formula (3) with $n=4$, we get

$$\int \cos^4 x\,dx = \frac{1}{4}\cos^3 x \sin x + \frac{3}{4}\int \cos^2 x\,dx$$

Again, apply Formula (3) with $n=2$, we get

$$\int \cos^4 x\,dx = \frac{1}{4}\cos^3 x \sin x + \frac{3}{4}\left(\frac{1}{2}\cos x \sin x + \frac{1}{2}\int dx\right)$$

$$= \frac{1}{4}\cos^3 x \sin x + \frac{3}{8}\cos x \sin x + \frac{3}{8}x + c$$

EXERCISE SET 7.2

4. Evaluate the integral $\int x^2 e^{-2x}\,dx$.

Solution: Using formula, we can write

$$\int x^2 e^{-2x}\,dx = x^2\int e^{-2x}\,dx - \int\{\frac{d}{dx}(x^2)\int e^{-2x}\,dx\}dx$$

$$= -\frac{1}{2}x^2e^{-2x} + \frac{1}{2}\int 2xe^{-2x}\,dx = -\frac{1}{2}x^2e^{-2x} + [x\int e^{-2x}\,dx + \frac{1}{2}\int e^{-2x}\,dx]$$

$$= -\frac{1}{2}x^2e^{-2x} - \frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + c$$

8. Evaluate the integral $\int x^2 \sin x\,dx$.

Solution: Using tabular integration by parts, we get

$$\int x^2 \sin x\,dx = -x^2\cos x + 2x\sin x + 2\cos x + c$$

10. Evaluate the integral $\int \sqrt{x}\ln x\,dx$.

Solution: Using formula, we get

$$\int \sqrt{x}\ln x\,dx = (\ln x)\left(\frac{2}{3}x^{3/2}\right) - \int \frac{1}{x}\times\frac{2}{3}x^{3/2}\,dx$$

$$= \frac{2}{3}x^{3/2}\ln x - \frac{2}{3}\times\frac{2}{3}x^{3/2} + c = \frac{2}{3}x^{3/2}\ln x - \frac{4}{9}x^{3/2} + c$$

11. Evaluate the integral $\int (\ln x)^2\,dx$.

Solution: Using formula, we get

$$\int (\ln x)^2\,dx = x(\ln x)^2 - 2\int \ln x\,dx$$

$$= x(\ln x)^2 - 2x\ln x + 2x + c$$

12. Evaluate the integral $\int \frac{\ln x}{\sqrt{x}}\,dx$.

Solution: Using formula, we get

$$\int \frac{\ln x}{\sqrt{x}}\,dx = 2\sqrt{x}\ln x - 2\int \frac{1}{\sqrt{x}}\,dx$$

$$= 2\sqrt{x}\ln x - 4\sqrt{x} + c$$

16. Evaluate the integral $\int \cos^{-1}(2x)\,dx.$

Solution: Using formula, we get

$$\int \cos^{-1}(2x)\,dx = x\cos^{-1}(2x) - \int \frac{-1}{\sqrt{1-4x^2}}(2x)dx$$

$$= x\cos^{-1}(2x) - \frac{1}{2}\sqrt{1-4x^2} + c$$

18. Evaluate the integral $\int x\tan^{-1}x\,dx.$

Solution: Using formula, we get

$$\int x\tan^{-1}x\,dx = \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int\frac{x^2}{1+x^2}dx = \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int\frac{(1+x^2)-1}{1+x^2}dx$$

$$= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int dx + \frac{1}{2}\int\frac{1}{1+x^2}dx = \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\tan^{-1}x + c$$

Formula: $\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + c$

20. Evaluate the integral $\int e^{3x}\cos 2x\,dx.$

Solution: Using formula, we get

$$\int e^{3x}\cos 2x\,dx = \frac{e^{3x}}{3^2+2^2}[3\cos 2x + 2\sin 2x] + c$$

$$= \frac{e^{3x}}{13}[3\cos 2x + 2\sin 2x] + c$$

21. Evaluate the integral $\int \sin(\ln x)\,dx.$

Solution: Let $I = \int \sin(\ln x)\,dx.$ Using formula, we get

$$I = \int \sin(\ln x)\,dx = x\sin(\ln x) - \int\frac{1}{x}\cos(\ln x)(x)\,dx = x\sin(\ln x) - \int\cos(\ln x)\,dx$$

$$\Rightarrow I = x\sin(\ln x) - x\cos(\ln x) - \int\frac{1}{x}\sin(\ln x)(x)\,dx$$

$$\Rightarrow I = x\sin(\ln x) - x\cos(\ln x) - I + c$$

Hence, $\int \sin(\ln x)\,dx = \frac{1}{2}x[\sin(\ln x) - \cos(\ln x)] + k.$

26. Evaluate the integral $\int \frac{xe^x}{(x+1)^2}dx.$

Solution: Using formula, we get

$$\int\frac{xe^x}{(x+1)^2}dx = (xe^x)\left(-\frac{1}{x+1}\right) - \int(xe^x + e^x)\left(-\frac{1}{x+1}\right)dx$$

$$= -\frac{xe^x}{x+1} + \int e^x dx = -\frac{xe^x}{x+1} + e^x + c$$

34. Evaluate the integral $\int_1^2 x\sec^{-1} x\,dx.$

Solution: Using formula, we get

$$\int x\sec^{-1} x\,dx = \frac{x^2}{2}\sec^{-1} x - \int \frac{1}{x\sqrt{x^2-1}}\left(\frac{x^2}{2}\right)dx$$

$$= \frac{x^2}{2}\sec^{-1} x - \frac{1}{2}\int \frac{x}{\sqrt{x^2-1}}\,dx = \frac{x^2}{2}\sec^{-1} x - \frac{1}{2}\sqrt{x^2-1}$$

Hence, $\int_1^2 x\sec^{-1} x\,dx = \frac{1}{2}\left[x^2\sec^{-1} x\right]_1^2 - \frac{1}{2}\left[\sqrt{x^2-1}\right]_1^2$

$$= \frac{1}{2}\left(\frac{4\pi}{3}-0\right) - \frac{1}{2}(\sqrt{3}-0) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

37. Evaluate the integral $\int_1^3 \sqrt{x}\tan^{-1}\sqrt{x}\,dx.$

Solution: Using formula, we get

$$\int \sqrt{x}\tan^{-1}\sqrt{x}\,dx = \frac{2}{3}\left(x^{3/2}\tan^{-1}\sqrt{x}\right) - \frac{1}{3}x + \frac{1}{3}\ln|1+x|$$

Hence, $\int_1^3 \sqrt{x}\tan^{-1}\sqrt{x}\,dx = \frac{2}{3}\left[x^{3/2}\tan^{-1}\sqrt{x}\right]_1^3 - \frac{1}{3}[x]_1^3 + \frac{1}{3}\left[\ln|1+x|\right]_1^3$

$$= \frac{2}{3}\left[3\sqrt{3}\times\frac{\pi}{3} - \frac{\pi}{4}\right] - \frac{1}{3}\times 2 + \frac{1}{3}[\ln 4 - \ln 2]$$

$$= \frac{1}{3}\left[2\sqrt{3}\,\pi - \frac{\pi}{2} - 2 + \ln 2\right].$$

Evaluate the integral using tabular integration by parts (47-52).

47. Evaluate the integral $\int (3x^2 - x + 2)e^{-x}dx.$

Solution: Using tabular integration by parts, we get

$$\int (3x^2-x+2)e^{-x}dx = -(3x^2-x+2)e^{-x} - (6x-1)e^{-x} - 6e^{-x} + c$$

$$= -(3x^2+5x+7)e^{-x} + c.$$

Repeated differentiation	**Repeated integration**	**Product of the expressions**	**Multiply the product by**
$3x^2-x+2$	e^{-x}		
$6x-1$	$-e^{-x}$	$(3x^2-x+2)(-e^{-x})$	$+1$
6	e^{-x}	$(6x-1)(e^{-x})$	-1
0	$-e^{-x}$	$6(-e^{-x})$	$+1$

48. Evaluate the integral $\int (x^2+x+1)\sin x\,dx.$

Solution: Using tabular integration by parts, we get

$$\int (x^2+x+1)\sin x\,dx = -(x^2+x+1)\cos x + (2x+1)\sin x + 2\cos x + c$$

$$= -(x^2+x-1)\cos x + (2x+1)\sin x + c$$

50. Evaluate the integral $\int x^3\sqrt{2x+1}\,dx.$

Solution: Using tabular integration by parts, we get

$$\int x^3\sqrt{2x+1}\,dx = \frac{1}{3}x^3(2x+1)^{3/2} - \frac{1}{5}x^2(2x+1)^{5/2} + \frac{2}{35}x(2x+1)^{7/2} - \frac{2}{315}(2x+1)^{9/2} + c.$$

54. (a) Evaluate the integral $\int_0^1 \frac{x^3}{\sqrt{x^2+1}}\,dx.$

Solution: Using formula, we get

$$\int \frac{x^3}{\sqrt{x^2+1}}\,dx = x^2\sqrt{x^2+1} - 2\int x\sqrt{x^2+1}\,dx = x^2\sqrt{x^2+1} - \frac{2}{3}(x^2+1)^{3/2}$$

Hence, $\int_0^1 \frac{x^3}{\sqrt{x^2+1}}\,dx = \sqrt{2} - \frac{2}{3}\times 2\sqrt{2} + \frac{2}{3} = \frac{2}{3} - \frac{\sqrt{2}}{3} = \frac{2-\sqrt{2}}{3}.$

55. (a) Find the area of the region enclosed by $y=\ln x$, the line $x=e$ and the x-axis.

Solution: Using formula, we get

$$\int_1^e \ln x\ dx = \left[x\ln x - x\right]_1^e = e\ln e - e - 0 + 1 = 1$$

61. Use reduction formula to evaluate (a) $\int \sin^4 x\,dx$, (b) $\int_0^{\pi/2} \sin^5 x\,dx.$

Solution: (a) Using Formula (4) with $m=4$, we get

$$\int \sin^4 x\,dx = -\frac{1}{4}\sin^3 x\cos x + \frac{3}{4}\int \sin^2 x\,dx$$

Again, apply Formula (4) with $m=2$, we get

$$\int \sin^4 x\,dx = -\frac{1}{4}\sin^3 x\cos x + \frac{3}{4}\left(-\frac{1}{2}\sin x\cos x + \frac{1}{2}\int dx\right)$$

$$= -\frac{1}{4}\sin^3 x\cos x - \frac{3}{8}\sin x\cos x + \frac{3}{8}x + c$$

(b) Applying reduction formula, we can find

$$\int \sin^5 x\,dx = -\frac{1}{5}\sin^4 x\cos x + \frac{4}{5}\int \sin^3 x\,dx$$

$$= -\frac{1}{5}\sin^4 x\cos x + \frac{4}{5}\left(-\frac{1}{3}\sin^2 x\cos x + \frac{2}{3}(-\cos x)\right)$$

$$= -\frac{1}{5}\sin^4 x\cos x - \frac{4}{15}\sin^2 x\cos x - \frac{8}{15}\cos x$$

Hence, $\int_0^{\pi/2} \sin^5 x\,dx = -\frac{1}{5}\left[\sin^4 x\cos x\right]_0^{\pi/2} - \frac{4}{15}\left[\sin^2 x\cos x\right]_0^{\pi/2} - \frac{8}{15}\left[\cos x\right]_0^{\pi/2} = \frac{8}{15}$